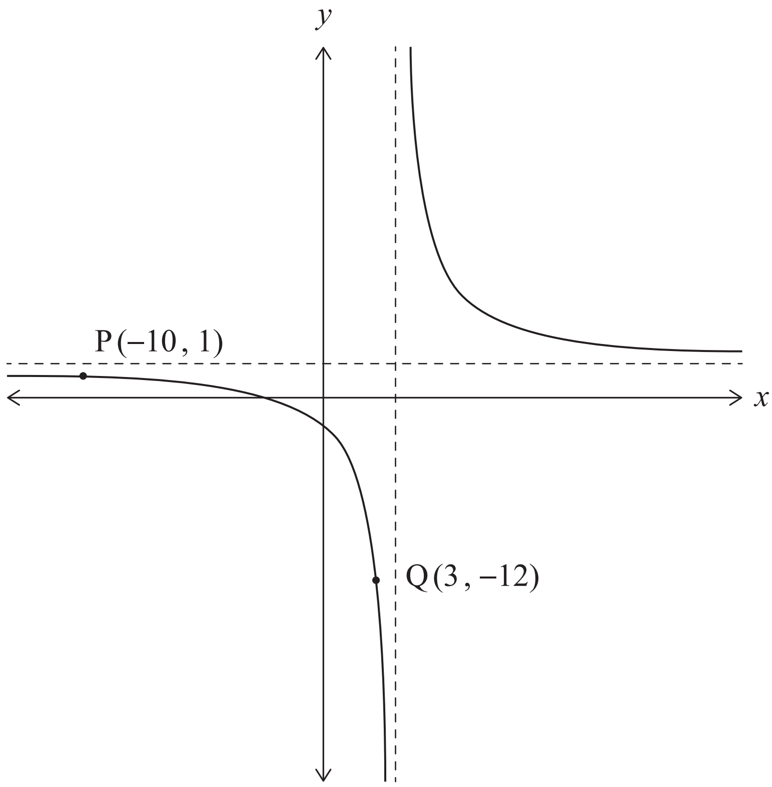
3. [Maximum mark: 8]

The following diagram shows the graph of $y = \dfrac{Ax + B}{x - 4}$, where $x \in \mathbb{R}$, $x \neq 4$ and $A, B \in \mathbb{Z}$.

The graph passes through the points $\mathrm{P}(-10, 1)$ and $\mathrm{Q}(3, -12)$.



(a) Determine the value of A and the value of B. [3]

(b) Describe a sequence of transformations that would map the graph of $y = \dfrac{1}{x}$ onto the graph of $y = \dfrac{Ax + B}{x - 4}$. [5]

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